



AI Resume Analyzer

Project Based Learning – Web Development

Academic Year
2025–2026



Abstract

The AI Resume Analyzer is a web-based application that evaluates resumes and provides a score along with improvement suggestions. Users upload their resume in PDF or TXT format, and the system extracts the text using JavaScript and analyzes it for important elements such as skills, projects, experience, certifications, and academic performance. The system calculates a resume score and generates professional suggestions to help improve the resume quality. This tool helps students and job seekers optimize their resumes for better career opportunities.



System Architecture

- 1 Users → Job Seekers / Students
- 2 Frontend → Resume Upload Interface (HTML, CSS)
- 3 Data Processing → Resume Text Extraction (PDF.js)
- 4 Analysis Engine → JavaScript-based Resume Scoring Algorithm
- 5 Output → Resume Score, Star Rating, and Improvement Suggestions



Key Features

- ✓ Resume Upload (PDF / TXT)
- ✓ Automatic Resume Scoring
- ✓ AI-based Suggestions
- ✓ Skill Detection System
- ✓ Star Rating Evaluation



Implementation – Learning Modules



CO1 Internet Fundamentals & HTML

- Resume upload interface
- Login page design
- Structured webpage layout
- Semantic HTML elements



CO2 CSS Design

- Responsive UI design
- Dashboard styling
- Card-based result display
- Flexbox layout



CO3 JavaScript Programming

- Login authentication logic
- Resume file upload handling
- PDF text extraction using PDF.js
- Resume scoring algorithm



CO4 JavaScript Interactivity

- Dynamic resume analysis
- Star rating generation
- Suggestion generation system
- Interactive dashboard updates



Conclusion & Future Scope

The AI Resume Analyzer demonstrates how web technologies can be used to analyze resumes and provide valuable feedback to users. The system automatically evaluates resumes and suggests improvements, helping users create stronger and more professional resumes. Future enhancements may include machine learning-based analysis, ATS compatibility checking, and integration with job recommendation systems.

